

AMBO UNIVERSITY

AMBO UNIVERSITY SMART CARD MANAGEMENT SYSTEM

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APPROVAL SHEET

This project report entitled "Ambo University Smart Card Management System" has been submitted to thefor examination and has been reviewed and approved by the assigned advisor and examiners.

DECLARATION

I hereby declare that this project entitled "Ambo University Smart Card Management System" is my own work and has been prepared for the academic requirements of the Department of Information Technology. All sources used in preparing this report are appropriately acknowledged.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to God for giving me the strength, knowledge, and perseverance to complete this project.

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Finally, I would like to thank my family and friends for their encouragement and support during my university study and throughout the development of this project.

ABSTRACT

The Ambo University Smart Card Management System is a web-based digital system developed to improve student registration, student information management, smart identification card generation, attendance tracking, and reporting.

The system uses React.js for the frontend, Node.js with Express.js for the backend, and MySQL for database management. Authentication and authorization are implemented using JWT-based authentication and role-based access control.

The system provides separate access for administrators, staff, and students. Administrators and staff can manage student information, approve registrations, generate smart cards, manage attendance, and generate reports, while students can manage their profiles, request smart cards, and view their own card and attendance information.

The Smart ID card includes student identification information, photograph, QR code, barcode, university branding, emergency contact information, and security elements. Cards can be exported as high-quality PNG or PDF files and prepared for printing.

Keywords: Smart Card, Student Management, React.js, Node.js, Express.js, MySQL, QR Code, Barcode, Attendance Management, Ambo University.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
DB	Database
JWT	JSON Web Token
QR	Quick Response
UI	User Interface
UX	User Experience
PDF	Portable Document Format
PNG	Portable Network Graphics
CRUD	Create, Read, Update, Delete
SQL	Structured Query Language
HTTP	Hypertext Transfer Protocol
MUI	Material UI

CHAPTER ONE — INTRODUCTION

1.1 Background of the Study

Reliable identification of students is central to how a university manages access, records attendance, and delivers services. Traditionally, student identification has depended on manually issued paper or laminated cards, supported by information kept in registers or disconnected spreadsheets. As student numbers grow, this manual approach becomes increasingly difficult to sustain.

Common problems with traditional student ID processes include manual registration, difficulty maintaining accurate student information, manual card preparation, the risk of duplicate card generation, difficult attendance tracking, slow report preparation, and limited accessibility of student data across departments.

The Ambo University Smart Card Management System was developed to address these problems. It is a web-based platform that digitizes student registration, centralizes student information, automates the generation of secure Smart ID cards, and supports attendance tracking and reporting through a single, role-based system.

1.2 Statement of the Problem

The existing, largely manual process of managing student identification at the university presents several challenges:

- Student information is maintained manually, increasing the risk of errors and inconsistency.
- Smart cards require manual preparation, which is slow and difficult to standardize.
- Tracking card issuance is difficult without a centralized record.
- Attendance records are difficult to manage and consolidate.
- Generating statistics and reports takes significant time and effort.
- Duplicate smart cards may be produced without proper validation against existing records.

1.3 General Objective

The general objective of this project is to design and develop a web-based Ambo University Smart Card Management System that improves student registration, identification-card generation, attendance management, and reporting.

1.4 Specific Objectives

The system aims to:

- Develop a student registration system.
- Manage student profiles.
- Implement role-based access control.
- Generate unique Smart ID cards.
- Generate QR codes and barcodes.
- Prevent duplicate card generation.
- Track attendance.

- Generate student and attendance reports.
- Provide dashboard statistics.
- Export Smart ID cards as PNG and PDF.
- Provide print-ready card output.
- Improve student information management.

1.5 Research Questions

- How can student registration be digitized?
- How can Smart ID cards be generated automatically?
- How can duplicate card generation be prevented?
- How can student attendance be managed digitally?
- How can administrators generate reports efficiently?
- How can role-based access improve system security?

1.6 Scope of the Project

The project covers student registration, student profile management, user authentication, role management, Smart ID generation, QR code generation, barcode generation, attendance management, card status management, reports, dashboard statistics, and PNG/PDF card export. The system is implemented with three user roles: Admin, Staff, and Student.

1.7 Significance of the Project

Students

- Easy registration
- Digital profile
- Smart ID access
- Attendance information

Staff

- Easier student management
- Card generation
- Attendance management
- Report generation

Administrators

- Centralized management
- Dashboard statistics
- Security and authorization
- Reports and analytics

University

- Reduced paperwork
- Better data organization

- Improved identification
- Faster information retrieval

1.8 Limitations of the Project

- The system currently focuses on student Smart ID management.
- Attendance currently depends on manual recording through the system.
- NFC functionality is represented as a card-design/security element rather than a fully implemented NFC reader system.
- The system requires a configured Node.js and MySQL environment.

CHAPTER TWO — LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing approaches to student identification and management, and the technologies adopted for the design and implementation of the Ambo University Smart Card Management System.

2.2 Student Identification Systems

Traditional student identification relies on manually issued cards supported by paper or spreadsheet-based records. Digital identification systems replace this with centralized databases, automated card generation, and machine-readable identifiers, improving accuracy, speed, and security.

2.3 Smart Card Technology

Smart identification cards combine visual identity information with machine-readable elements such as QR codes and barcodes that allow rapid digital verification. Security features such as unique card identifiers and hologram-style design elements help reduce the risk of forgery.

2.4 Student Management Systems

Universities increasingly rely on computerized student management systems to maintain accurate, centralized student records, streamline registration, and support administrative decision-making.

2.5 Attendance Management Systems

Computerized attendance tracking replaces manual roll-calls with digital records that can be aggregated, searched, and reported far more efficiently than paper-based logs.

2.6 Web-Based Information Systems

Web-based systems provide centralized access from any device with a browser, simplify deployment and maintenance, and make it easier to enforce consistent business rules and role-based access across an organization.

2.7 Technologies Used

The system was built using React.js and Material-UI for the frontend, with data fetched through the Fetch API. Client-side QR code generation, JsBarcode for barcode generation, and html2canvas with jsPDF for image and PDF export are used to produce the Smart ID cards. The backend is implemented with Node.js and Express.js, backed by a MySQL database, with JWT and bcrypt used for authentication and password security.

CHAPTER THREE — SYSTEM ANALYSIS

3.1 Existing System

Before the proposed system, student registration, identification, attendance, and reporting were handled through a combination of manual paper forms, locally prepared ID cards, and disconnected records maintained by different offices.

3.2 Problems of Existing System

- Manual work
- Paper-based records
- Data duplication
- Slow report generation
- Difficult card management
- Limited security

3.3 Proposed System

The proposed Ambo University Smart Card Management System digitizes registration, centralizes student data in a MySQL database, automates Smart ID card generation with duplicate prevention, and provides role-based dashboards for attendance and reporting.

3.4 Functional Requirements

The system should allow:

- Registration
- Login
- Profile completion
- Student management
- Approval/rejection
- Smart card generation
- Card blocking/unblocking
- Attendance management
- Reports
- Card printing/export

3.5 Non-Functional Requirements

Security

JWT authentication and role-based authorization protect system resources and restrict actions according to user role.

Performance

The system is designed for fast database queries and a responsive user interface.

Usability

A simple, consistent user interface supports ease of use for Admin, Staff, and Student roles.

Reliability

Persistent, database-backed storage ensures information is not lost between sessions.

Maintainability

A clear separation between the React frontend and Express backend supports easier maintenance and future extension.

CHAPTER FOUR — SYSTEM DESIGN

4.1 System Architecture

The system follows a three-tier architecture: React.js (frontend) → Node.js/Express.js (backend) → MySQL (database). The documented ports are frontend 3007, backend 5000, and MySQL 3306.

4.2 Use Case Diagram

Actors: Admin, Staff, Student.

Use cases: Login, Register, Manage Students, Approve Registration, Generate Smart Card, Block Card, Mark Attendance, View Reports, Manage Profile.

4.3 Data Flow Diagram

User → React Frontend → Express API → MySQL → API Response → React UI.

4.4 Entity Relationship Diagram

The documented database schema defines four main tables: users, students, cards, and attendance.

4.5 Database Design

Each table stores a distinct part of the system's data: the users table holds authentication and role information; the students table holds personal, academic, and contact details; the cards table holds Smart ID card records linked to a student, including card status and generation metadata; and the attendance table records date, time, and status entries linked to a student.

4.6 Role-Based Access Design

Access to system functions is governed by an Admin/Staff/Student permission matrix, ensuring that each role can only perform the actions appropriate to it — for example, only Admin and Staff can approve registrations or generate cards, while Students can only manage their own profile and view their own card and attendance information.

CHAPTER FIVE — SYSTEM IMPLEMENTATION

5.1 Development Environment

- Node.js
- React.js
- Express.js
- MySQL
- VS Code
- XAMPP/phpMyAdmin (for MySQL management)
- Web Browser

5.2 Backend Implementation

The Express backend exposes authentication, student, registration, card, dashboard, and attendance routes. The documented API structure includes `/api/auth`, `/api/students`, `/api/registrations`, `/api/cards`, `/api/dashboard`, and `/api/attendance`.

5.3 Frontend Implementation

The React frontend provides pages/components for Login, Dashboard, Student Registration, Students, Registrations, Smart Card Generator, Cards, Attendance, Reports, and a shared Layout.

5.4 Authentication

The authentication flow proceeds as follows:

1. The user enters credentials.
2. The backend verifies the credentials.
3. A JWT token is generated.
4. The token is sent with subsequent protected requests.
5. Middleware verifies the token on each request.
6. Role permissions are checked before the requested action is performed.

CHAPTER SIX — SMART ID CARD IMPLEMENTATION

6.1 Smart Card Generation

The card generation process proceeds as follows: Approved Student → Generate Card → Check Existing Card → Generate Unique Card ID → Generate QR/Barcode → Save Card → Display Card. The system prevents duplicate generation and returns a 409 Conflict response when a card already exists for a student.

6.2 Front Side

- Ambo University logo
- University header
- Student Identification Card title
- QR code
- University information
- Signature

6.3 Back Side

- Student name
- Student ID
- Department
- Program
- Email
- Emergency contact
- Photo
- Barcode
- Security hologram

6.4 Card Size

The Smart ID card is produced at 8.7 cm × 5.5 cm, rendered at 300 DPI, giving 1028 × 650 pixels per side, and exported as PNG or PDF.

6.5 Front and Back Side-by-Side Printing

For the combined export, the front and back sides are arranged side-by-side at 300-DPI output for the PNG export, and placed together on one page for the PDF export, ready for the print dialog.

CHAPTER SEVEN — SYSTEM TESTING

7.1 Testing Introduction

This chapter describes the testing carried out to verify that the system functions correctly and meets its stated requirements, covering unit, integration, authentication, registration, Smart Card, attendance, and report testing.

7.2 Unit Testing

Individual functions and components were tested in isolation to confirm they behave as expected.

7.3 Integration Testing

End-to-end integration testing verified the flow React → API → MySQL, confirming that data entered through the frontend is correctly processed by the backend and persisted in the database.

7.4 Authentication Testing

- Valid login
- Invalid login
- Unauthorized request
- Expired/invalid token
- Role restrictions

7.5 Registration Testing

- Student registration
- Profile completion
- Approval
- Rejection
- Duplicate student ID

7.6 Smart Card Testing

- Card generation
- Duplicate card prevention
- QR code
- Barcode
- Photo
- PDF
- PNG
- Front/back output

7.7 Attendance Testing

- Mark attendance
- Date

- Time
- Present
- Absent
- Late

7.8 Report Testing

- Student report
- Attendance report
- Card report
- Department report
- Year report
- CSV export
- Print

CHAPTER EIGHT — SYSTEM RESULTS AND DISCUSSION

8.1 Login Result



YUUNIVARSITI
AMBOO
አምቦ ዩኒቨርሲቲ | AMBO UNIVERSITY

Smart Card Management System

Sign in to your account

Enter your password

LOGIN

OR

New Student?

REGISTER HERE

8.2 Dashboard Result

The dashboard provides total students, total cards issued, pending registrations, and today's attendance. [

 Smart Card Management System

 admin  LOGOUT

]

8.3 Student Registration Result

Student Account Creation

Create your account to get started

Minimum 8 characters

CREATE ACCOUNT

Already have an account? [Login here](#)

[

]

8.4 Student Management Result

Smart Card Management System

Y Yonas Guteta LOGOUT

Welcome, Yonas Guteta!

REFRESH PROFILE

My Profile



Name: **Yonas Guteta**
Student ID: **2026-2266**
Department: **Computer Science**
Year: **Year 4**
Phone: **0965784143**
Status: **Profile Complete**

EDIT PROFILE

Quick Links

- View your attendance records
- Download your card details
- Update your profile information

8.5 Smart Card Generation Result


AMBO UNIVERSITY
STUDENT IDENTIFICATION CARD

P.O. Box: 19
Tel No: 251-112-382215
Fax No: 251-112-382037
Ambo.mbo.edu.et
Date of Issue: December 2024

DIGITAL VERIFICATION CODE


registrar@ambou.edu.et
Student ID Card Use Policy
1. While enrolled: Right to use the Smart ID and university services; obligation to protect and use it properly.
2. After graduation: Right to use it for alumni identification/services; obligation not to use student-only privileges.
3. After withdrawal: Right to access authorized records; obligation to stop using the ID and return/deactivate it as required.
4. If the ID is lost: Report the loss immediately, deactivate/block the ID to prevent misuse, and request a replacement ID.

Authorized Signature

BACK SIDE - SMART I.D.


AMBO UNIVERSITY
REGULAR STUDENT S I.D CARD

Yonas Guteta
STUDENT ID: 2026-2266
DEPARTMENT: Computer Science
PROGRAM: Regular
johnguteta465@gmail.com
EMERGENCY: Guteta chimdi - 0960133808





SEC

FRONT SIDE - CARD DETAILS

8.7 Attendance Result

Attendance Reports

dd/mm/yyyy

[EXPORT CSV](#)

ID	Student ID	Name	Department	Year	Smart Card ID	Date	Time
6	2026-2266	Yonas Guteta	Computer Science	4	CARD-2026-0012	2026-08-19T21:00:00.000Z	17:18:09
5	2026-4795	Gemechu Abe	Information Systems	4	CARD-2026-0004	2026-08-18T21:00:00.000Z	22:53:08
4	2026-4795	Gemechu Abe	Information Systems	4	CARD-2026-0004	2026-08-18T21:00:00.000Z	21:33:14
3	2026-9324	Gemeda Chimdessa	Electrical Engineering	1	CARD-2026-0009	2026-08-18T21:00:00.000Z	21:30:25
2	2026-4795	Gemechu Abe	Information Systems	4	CARD-2026-0004	2026-08-18T21:00:00.000Z	21:11:44

8.10 Current System Statistics

According to the supplied documentation, the system records 9 approved students, 8 generated Smart Cards, and 1 student pending card generation, while attendance has not yet been recorded.

CHAPTER NINE — CONCLUSION AND RECOMMENDATIONS

9.1 Conclusion

The Ambo University Smart Card Management System successfully achieves its stated objectives. It replaces manual registration with a digital process, centralizes student information, automates Smart ID generation with QR/barcode identification and duplicate prevention, supports attendance tracking, and provides reporting through a role-based, secure platform capable of producing high-quality exportable card output.

9.2 Recommendations

The following are recommended as future improvements rather than features currently implemented:

- NFC-based card verification.
- QR-code scanning at university gates.
- Mobile application.
- Email/SMS notifications.
- Advanced analytics.
- Automated attendance scanning.
- Online card renewal.
- Card expiry management.
- Backup and recovery automation.
- Deployment to university production infrastructure.

REFERENCES

The following sources were consulted during the development of this project:

- React.js Documentation — <https://react.dev>
- Node.js Documentation — <https://nodejs.org/docs>
- Express.js Documentation — <https://expressjs.com>
- MySQL Documentation — <https://dev.mysql.com/doc>
- JSON Web Tokens (JWT) Documentation — <https://jwt.io>
- QR Code Library Documentation
- jsPDF Documentation

APPENDICES

Appendix A — Database SQL

- Database creation
- Tables
- Constraints
- Indexes

Appendix B — API Endpoints

Includes all API routes under /api/auth, /api/students, /api/registrations, /api/cards, /api/dashboard, and /api/attendance.

Appendix C — Screenshots

- 1. Login
- 2. Registration
- 3. Dashboard
- 4. Students
- 5. Registrations
- 6. Smart Card Generator
- 7. Cards
- 8. Attendance
- 9. Reports

Appendix D — Smart ID Designs

- Front side
- Back side
- Front + Back side-by-side
- PDF print version
- PNG print version

Appendix E — Sample Reports

- Student registration report
- Attendance report
- Card issuance report
- Department report
- Year-wise report

Appendix F — Source Code Structure

```
ambo-smart-card/  frontend/src/{components, pages, services, App.js}
backend/{routes, middleware, controllers, uploads, server.js}
database/ambo_smart_card.sql
```

Appendix G — User Manual

Admin

- 1. Login
- 2. Open dashboard
- 3. Manage registrations
- 4. Approve students
- 5. Generate cards
- 6. View cards
- 7. Manage attendance
- 8. Generate reports

Staff

- 1. Login
- 2. Manage students
- 3. Generate cards
- 4. Mark attendance
- 5. View reports

Student

- 1. Register
- 2. Login
- 3. Complete profile
- 4. Request Smart ID
- 5. View Smart ID
- 6. View own information/attendance